

Assignment-Regression Algorithm

1.) Identify your problem statement.

Requirement is clear. To, predict the insurance charges based on the given parameters like age, sex, smoker, BMI and no. of child.

2.) Tell basic info about the dataset (Total number of rows, columns).

Total no. of rows → 1338

Total no. of columns → 6

Independent inputs will be total 5.

3.) As part of preprocessing, Converting string column data type to the number as a nominal data. By using one hot encoding method.

Both Sex column and smoker column will be pre-processed.

1. MLR algorithm r_score == 0.7894790349867009

2. SVM ALGORITHM – R_score- Value

SVM Algorithm				
Hyper Parameter	Kernel			
C Value	linear(r_score)	rbf(r_score)	poly(r_score)	sigmoid(r_score)
1	-0.010102665	-0.083382386	-0.075699656	-0.075429243
10	0.462468414	-0.032273294	0.038716223	0.039307144
100	0.628879286	0.320031783	0.617956962	0.527610355
1000	0.764931174	0.810206485	0.617956962	0.287470695
2000	0.744041831	0.854776643	0.860557926	-0.593950973
3000	0.74142366	0.866339395	0.859893008	-2.124419479

Here, we can clearly see that SVM algorithm with input Kernel = Radial Bias Function(rbf) along with hyper parameter C = 3000 gives the best r_score validation result.

3. DECISION TREE – R_score – Value

DECISION TREE		
Crieterion	Splitter	R_Score
squared_error	Best	0.679766222
squared_error	Random	0.725579091
absolute_error	Best	0.653017823
absolute_error	Random	0.718656083
poisson	Best	0.730084846
poisson	Random	0.648920867

4. RANDOM FOREST- R_score – value

RANDOM FOREST		
Crieterion	n_estimators	R_Score
squared_error	10	0.833785568
squared_error	50	0.849978109
squared_error	100	0.853978132
absolute_error	10	0.848278752
absolute_error	50	0.848278752
absolute_error	100	0.849802036
poisson	10	0.831229967
poisson	50	0.848772101
poisson	100	0.85258863

As, part of the conclusion, SVMR algorithm gives the better performance.

And the finalized model is supervised learning, SVM – regression with input parameters
SVR(kernel='rbf', C = 3000).

SVM Algorithm

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<code>sigmoid(r_score)</code>
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0.527610355
0.287470695
-0.593950973
-2.124419479